

SRIRAM S U

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PROFILE SUMMARY

Software Engineer with close to 1 year of hands-on experience building industrial automation and IIoT-driven web and desktop applications using Angular, Node.js, and Electron. Specialized in developing file-based data ingestion pipelines, real-time data processing systems, and device integration using MQTT and serial communication. Experienced in designing event-driven architectures, background processing systems, and scalable automation workflows for industrial environments.

WORK EXPERIENCE

Full Stack Software Developer

Coimbatore, India

Acceedo

AUG 2025 - Present

- Designed and developed production-grade web and desktop applications using Angular, Node.js, Express.js, and MongoDB.
- Built workflow-driven systems with conditional state transitions, role-based access control, and approval flows.
- Developed and optimized RESTful APIs to support inspection data, automation workflows, and application state management.
- Implemented automation and background processing pipelines for file-based data ingestion, reducing manual effort and improving system reliability.
- Integrated RxJS, NgRx, and WebSockets to manage complex application state and enable real-time status updates.
- Collaborated with cross-functional teams and contributed to responsive UI development using Angular Material and Bootstrap.
- Implemented real-time data ingestion using MQTT protocol for machine-to-server communication.

PROJECTS

Desktop Automation Application

Tech: Angular (UI), Electron, Node.js, File System APIs, IPC, Serial Port, IP Communication

- Built a desktop automation system to ingest and process industrial data generated by melting software.
- Designed a file-based ingestion pipeline using directory monitoring and background processing for automated data extraction.
- Implemented event-driven workflows to transmit processed data over IP and serial interfaces with real-time status tracking.
- Optimized XML processing by extracting only required metadata, preventing UI blocking and improving performance.
- Used Electron IPC to separate UI rendering from backend file operations, ensuring responsiveness and reliability.

Quality Management System (Industry-Oriented Web Application)

Tech: Angular, Node.js, Express.js, MongoDB

- Developed a Quality Management System to digitize and automate shop-floor quality inspection workflows.
- Implemented initial and periodic inspection workflows with configurable rules to validate quality parameters.
- Designed dynamic Angular forms driven by product and process configurations.
- Implemented conditional state transitions to control production flow (continue, hold, resume).
- Built RESTful APIs and integrated MongoDB for inspection data storage, approvals, and audit traceability.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Java, SQL

Technologies & Tools: Angular, Node.js, Express.js, MongoDB, MySQL, JSON, Electron, RESTful APIs, RxJS, NgRx, WebSockets, Socket.io, Docker (containerization), Nginx (basic configuration), Git, GitHub, Postman

EDUCATION

Saranathan College of Engineering

Bachelor of Engineering

Major in Electronic and Communication Engineering (ECE) - CGPA: 7.88

Trichy, Tamil Nadu

May 2024